

CALL CENTER DATA ANALYSIS REPORT

EXCEL DATA ANALYSIS PROJECT

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Executive Summary

This report presents the findings of a comprehensive analysis of call center operations conducted by the Data Analytics Team for the period ending May 2025. Utilizing advanced Excel-based analytical tools — including Pivot Tables, KPI Cards, and an interactive Dashboard — the analysis evaluated a dataset of 1,000 customer call records to assess operational efficiency, representative performance, and customer satisfaction outcomes.

Key findings indicate that the call center achieved a total purchase conversion value of \$96,623 across all interactions, with a cumulative call duration of 89,850 minutes. Customer satisfaction metrics are broadly positive, with 30.7% of callers awarding the highest five-star rating and the majority of respondents providing ratings of four stars or above.

Representative R02 managed the highest call volume at 218 interactions, while Representative R03 led in revenue generation at approximately \$20,900. Saturday emerged as the peak operational day by call volume, with Thursday recording the lowest activity. Based on these findings, this report sets forth a series of strategic recommendations to enhance workforce efficiency, sustain customer satisfaction levels, and leverage data-driven insights for continuous performance improvement.

1. Introduction

The primary objective of this analytical initiative is to examine call center operational data with a view to identifying patterns and trends related to customer interactions, representative performance, customer satisfaction, and purchase behaviors. In an increasingly competitive service environment, the ability to derive actionable intelligence from operational data represents a significant strategic advantage for organizations seeking to optimize both customer experience and commercial outcomes.

This report has been produced by the Data Analytics Team as part of an ongoing commitment to evidence-based performance management. The analysis leverages Microsoft Excel as the primary analytical platform, employing a range of tools including Pivot Tables, Pivot Charts, Conditional Formatting, KPI Cards, and a fully interactive Dashboard to transform raw data into structured, executive-level insights.

The findings presented herein are intended to inform operational decision-making at both a strategic and a day-to-day management level. The report is structured to progress from a high-level executive summary through detailed findings to targeted recommendations, ensuring that readers at all levels of the organization can derive value from the content.

2. Dataset Overview

The dataset underpinning this analysis comprises 1,000 individual customer call records, each containing a rich set of attributes that facilitate multidimensional analysis. The fields captured within the dataset include customer identification information, assigned representative details, call duration in minutes, purchase amount in US dollars, customer satisfaction ratings, caller gender, city of origin, and the date and time of each interaction.

These data fields collectively enable an integrated assessment of operational performance across four principal dimensions: individual representative productivity, temporal call volume patterns, customer satisfaction distribution, and demographic segmentation by city and gender. The breadth and granularity of the dataset provide a robust foundation for the analytical findings and recommendations presented in the subsequent sections of this report.

3. Methodology

The analytical methodology employed in this project adheres to established best practices in data preparation, exploratory analysis, and visualization. Prior to analysis, the dataset underwent a thorough data quality review to ensure the integrity and reliability of all findings.

3.1 Data Preparation and Cleaning

The data preparation phase involved a systematic review of the raw dataset to identify and resolve any quality issues. Duplicate records were identified and removed to eliminate the risk of double-counting in aggregate calculations. Data types were verified across all fields, with particular attention paid to date fields, which were formatted consistently to enable accurate temporal analysis. Numerical fields, including purchase amounts and customer ratings, were validated against expected ranges, and categorical values — such as representative identifiers and city names — were standardized to ensure consistency throughout the dataset.

3.2 Analytical Tools and Techniques

Following data preparation, the cleansed dataset was subjected to analysis using a suite of Microsoft Excel features. Pivot Tables were constructed to aggregate and summarize call volumes, revenue figures, and satisfaction ratings across the key analytical dimensions. Pivot Charts were generated to provide visual representations of the underlying data distributions and trends. KPI Cards were configured to surface headline performance metrics at a glance, and an interactive Dashboard incorporating Slicers was developed to enable dynamic, user-driven exploration of the data. All analytical outputs were reviewed for accuracy prior to inclusion in this report.

4. Key Performance Indicators

The following table summarizes the principal key performance indicators (KPIs) derived from the analysis. Each metric is presented alongside a benchmark or target value and an assessment of current performance status. These indicators collectively provide a high-level view of operational health and serve as the foundation for the detailed findings discussed in Section 5.

Metric	Value	Benchmark / Target	Status
Total Calls Received	1,000	1,000 / month	✓ On Track
Total Purchase Amount	\$96,623	\$90,000 / month	✓ On Track
Total Call Duration	89,850 minutes	≤ 95,000 minutes	✓ On Track
Happy Callers (5-Star)	307 (30.7%)	≥ 30%	✓ On Track
Average Customer Rating	4.0+ Stars	≥ 4.0 Stars	✓ On Track
Top Representative (Revenue)	R03 — \$20,900	\$18,000 target	✓ On Track

As the foregoing table illustrates, the call center is performing at or above target across the majority of key metrics. The sole area identified for review relates to the distribution of call volumes across representatives, with R02 absorbing a disproportionately high share of total interactions. This matter is addressed in detail within the Representative Analysis subsection and in the Strategic Recommendations section of this report.

5. Detailed Findings

5.1 Representative Performance Analysis

An examination of individual representative performance reveals a measurable degree of variation in both call volume and revenue generation across the team. Representative R02 recorded the highest total call count for the period, handling 218 of the 1,000 calls logged — equivalent to 21.8% of total volume. This concentration of workload in a single representative warrants attention from a workforce planning perspective, as it may indicate an imbalance in call routing protocols or an inequitable distribution of working hours.

In contrast, Representative R03 demonstrated the strongest commercial performance, generating an estimated \$20,900 in purchase conversions — the highest revenue figure among all representatives for the period. This performance is indicative of strong customer engagement skills and effective consultative selling techniques. The disparity between call volume leadership (R02) and revenue leadership (R03) underscores the importance of evaluating representatives across multiple performance dimensions rather than relying solely on activity-based metrics.

5.2 Day-of-Week Call Volume Analysis

Analysis of call volume by day of the week reveals a clear pattern of demand concentration towards the end of the working week and into the weekend. Saturday recorded the highest aggregate call volume of any day in the period under review, suggesting that customers exhibit a strong preference for contacting the call center on weekend days — likely reflecting their own availability outside of standard working hours.

Conversely, Thursday registered the lowest call volume of any day in the dataset. This finding presents a potential operational opportunity: the excess capacity available on Thursdays could be leveraged for representative training, quality assurance reviews, or process improvement activities without adversely affecting service levels. Correspondingly, Saturday staffing levels should be reviewed to ensure that adequate resources are deployed to meet peak demand and maintain acceptable call response times.

5.3 Customer Satisfaction Analysis

Customer satisfaction data for the period indicates a broadly positive service experience across the caller population. The majority of respondents provided ratings of either four or five stars, reflecting a high level of satisfaction with the service received. Specifically, 307 callers — representing 30.7% of the total sample — awarded the maximum five-star rating, a figure that meets the organization's established benchmark for premium service delivery.

Whilst the overall satisfaction picture is encouraging, the presence of lower-rated interactions suggests that there remains scope for improvement, particularly in addressing the service factors that give rise to sub-four-star ratings. A targeted quality assurance initiative, focused on the specific interaction characteristics correlated with lower satisfaction scores, would be a productive means of incrementally improving the

overall rating distribution and strengthening the organizations' customer experience proposition.

5.4 City and Gender Demographic Analysis

The geographic and demographic breakdown of call data reveals distinct patterns across the cities represented in the dataset. Cleveland recorded the highest volume of female callers during the period, while Columbus registered the greatest number of male callers. These demographic variations may reflect differences in the underlying customer base in each city or divergent patterns in product and service usage by gender.

These insights carry practical implications for both marketing and customer experience strategy. Tailoring communication styles, product recommendations, and outreach campaigns to reflect the demographic profile of callers in each city presents an opportunity to enhance relevance and personalization in customer interactions. Further segmentation analysis, incorporating purchasing behaviors by demographic group, would yield additional strategic value and is recommended as a follow-on analytical activity.

6. Strategic Recommendations

Drawing upon the findings presented in Section 5, the Data Analytics Team sets forth the following seven strategic recommendations. These recommendations are prioritized and assigned to responsible ownership functions to facilitate implementation planning and accountability. The table below presents a structured overview of each recommended action.

Action Item	Priority	Owner	Timeline
Recognized and incentivize top-performing representatives, particularly R02 and R03, to sustain high performance levels.	High	Operations Manager	Immediate
Implement targeted coaching and skills development for representatives with below-average conversion and satisfaction metrics.	High	HR & Training Lead	30 Days
Increase staffing levels and shift coverage on Saturdays to address peak call volumes and reduce wait times.	High	Workforce Planning	30 Days
Deploy workload-balancing protocols to distribute call volumes equitably across all representatives.	Medium	Operations Manager	60 Days
Launch a customer satisfaction improvement initiative targeting callers who rated service below 4 stars.	Medium	Quality Assurance	60 Days
Develop gender- and city-specific engagement strategies for Cleveland and Columbus based on demographic call patterns.	Low	Marketing & CX Team	90 Days
Establish a real-time KPI monitoring dashboard with weekly review cadence to track performance continuously.	Medium	Data Analytics Team	60 Days

The seven recommendations outlined above collectively address the principal improvement opportunities identified through the analysis. High-priority actions — encompassing representative recognition, targeted training, and Saturday staffing — should be progressed immediately to deliver near-term performance improvements. Medium- and low-priority actions represent longer-horizon investments in operational resilience, customer experience quality, and strategic market engagement that will compound in value over time.

7. Conclusion

This report has presented a comprehensive, data-driven analysis of call center operations for the period ending May 2025. The findings demonstrate that the organization is performing at a strong level across the majority of its key performance indicators, with total purchase conversions of \$96,623, a call volume of 1,000 interactions, and a customer satisfaction profile in which over 30% of callers awarded the highest available rating.

At the same time, the analysis has surfaced a number of specific opportunities for operational improvement. The uneven distribution of call volumes across representatives, the concentration of demand on Saturdays, and the presence of below-benchmark satisfaction ratings in a subset of interactions are all areas in which targeted management action can yield meaningful improvements in efficiency, service quality, and commercial performance.

The interactive Excel Dashboard developed as part of this project provides the organization with a powerful and flexible tool for ongoing performance monitoring. By embedding a culture of regular KPI review and data-driven decision-making — supported by the dashboard infrastructure now in place — the call center is well-positioned to sustain its current strengths, address its identified gaps, and continue to deliver high-quality service outcomes for its customers.

The Data Analytics Team remains committed to providing analytical support to operational and strategic management functions, and welcomes engagement on any aspect of the findings or recommendations presented in this report.

— End of Report —
